

## IXOPO, South Africa

WMO: 685750

Lat: 30.1525S

Lon: 30.0747E

Elev: 3074

StdP: 13.14

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 7             | 36.0       | 37.9 | 20.6                          | 17.3 | 57.9 | 25.1 | 21.5 | 58.2 | 23.8                  | 55.9 | 18.7 | 57.5 | 3.4                      | 0    | 0.332 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 15.9                   | 89.6            | 67.6 | 86.4 | 67.6 | 83.2 | 67.2 | 73.2                | 83.0 | 71.7 | 80.8 | 70.4 | 78.5 | 4.9                  | 90   |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 70.2                            | 124.9 | 78.0 | 69.0 | 119.7 | 76.3 | 67.9 | 115.1 | 74.7 | 39.1          | 83.3 | 37.7 | 81.1 | 36.5 | 78.7 | 78.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |    |    |
|-------------------|------|-----|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|----|----|
|                   |      |     | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |    |    |
| (4) 13.3          | 11.1 | 9.3 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 62.6   | 69.7  | 70.2  | 68.5  | 64.2  | 59.2  | 55.0  | 53.9  | 57.1  | 60.6  | 62.6  | 64.2  | 66.9  |
|                                                                        | DBStd       | 7.64   | 5.06  | 4.85  | 5.09  | 5.38  | 4.87  | 4.07  | 4.83  | 5.35  | 6.51  | 6.46  | 6.14  | 5.94  |
|                                                                        | HDD50       | 36     | 0     | 0     | 0     | 0     | 2     | 7     | 18    | 5     | 4     | 0     | 0     | 0     |
|                                                                        | HDD65       | 1639   | 14    | 10    | 27    | 78    | 187   | 301   | 346   | 251   | 161   | 125   | 89    | 50    |
|                                                                        | CDD50       | 4644   | 612   | 567   | 573   | 426   | 286   | 156   | 138   | 226   | 321   | 390   | 426   | 523   |
|                                                                        | CDD65       | 772    | 162   | 157   | 134   | 54    | 6     | 0     | 1     | 7     | 29    | 49    | 65    | 108   |
|                                                                        | CDH74       | 4734   | 866   | 725   | 680   | 297   | 107   | 9     | 16    | 133   | 319   | 449   | 480   | 653   |
|                                                                        | CDH80       | 1410   | 287   | 217   | 193   | 63    | 14    | 0     | 34    | 94    | 149   | 159   | 200   | 200   |
| (14) Wind                                                              | WSAvg       | 3.2    | 2.9   | 3.1   | 2.7   | 2.8   | 3.0   | 3.5   | 4.1   | 3.2   | 3.3   | 3.4   | 3.4   | 3.5   |
| (15) Precipitation                                                     | PrecAvg     | 30.7   | 4.8   | 3.5   | 3.7   | 1.9   | 0.7   | 0.8   | 1.0   | 1.1   | 1.7   | 3.5   | 3.5   | 4.8   |
|                                                                        | PrecMax     | 40.9   | 8.7   | 7.4   | 6.8   | 4.7   | 2.8   | 4.6   | 7.5   | 3.6   | 6.9   | 6.5   | 7.5   | 12.0  |
|                                                                        | PrecMin     | 19.3   | 1.4   | 1.7   | 1.6   | 0.2   | 0.0   | 0.0   | 0.0   | 0.0   | 0.6   | 1.1   | 1.8   | 1.8   |
|                                                                        | PrecStd     | 6.1    | 1.7   | 1.3   | 1.6   | 1.1   | 0.7   | 1.1   | 1.6   | 1.0   | 1.4   | 1.6   | 1.4   | 2.3   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 92.0  | 91.3  | 89.9  | 86.1  | 82.3  | 74.7  | 76.0  | 84.4  | 89.7  | 90.9  | 91.3  | 91.5  |
|                                                                        |             | MCWB   | 71.1  | 71.6  | 67.8  | 64.5  | 58.0  | 53.6  | 52.9  | 56.9  | 60.0  | 62.6  | 65.5  | 71.4  |
|                                                                        | 2%          | DB     | 87.0  | 86.1  | 85.4  | 81.0  | 77.3  | 70.9  | 71.0  | 77.7  | 82.8  | 84.9  | 85.0  | 85.4  |
|                                                                        |             | MCWB   | 71.3  | 71.4  | 68.3  | 64.1  | 57.9  | 52.3  | 51.3  | 55.2  | 60.1  | 63.0  | 67.2  | 71.3  |
|                                                                        | 5%          | DB     | 82.7  | 81.9  | 81.0  | 77.1  | 73.1  | 67.5  | 67.4  | 72.6  | 77.5  | 79.6  | 79.9  | 81.0  |
|                                                                        |             | MCWB   | 70.5  | 70.7  | 68.3  | 63.7  | 57.9  | 51.8  | 50.6  | 55.5  | 59.9  | 63.5  | 66.4  | 69.9  |
|                                                                        | 10%         | DB     | 78.6  | 78.0  | 77.4  | 73.1  | 69.1  | 64.1  | 63.8  | 68.0  | 72.6  | 74.5  | 75.3  | 76.9  |
|                                                                        |             | MCWB   | 69.7  | 69.2  | 67.4  | 62.7  | 57.2  | 50.7  | 50.0  | 54.5  | 59.0  | 62.6  | 65.2  | 68.0  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 75.2  | 75.1  | 72.6  | 69.8  | 63.8  | 58.9  | 56.9  | 62.4  | 65.9  | 68.4  | 72.2  | 74.8  |
|                                                                        |             | MCDB   | 84.3  | 86.3  | 82.4  | 79.4  | 71.2  | 65.8  | 65.4  | 71.0  | 80.4  | 79.8  | 83.1  | 85.3  |
|                                                                        | 2%          | WB     | 72.9  | 72.6  | 70.7  | 66.8  | 61.5  | 56.1  | 55.4  | 60.1  | 63.3  | 66.5  | 69.4  | 72.5  |
|                                                                        |             | MCDB   | 82.2  | 82.1  | 79.3  | 75.1  | 70.1  | 63.2  | 63.0  | 70.0  | 74.7  | 77.7  | 78.4  | 82.2  |
|                                                                        | 5%          | WB     | 71.4  | 71.1  | 69.2  | 65.3  | 59.9  | 54.2  | 53.6  | 57.8  | 61.7  | 64.8  | 67.4  | 70.6  |
|                                                                        |             | MCDB   | 80.2  | 79.0  | 77.0  | 72.7  | 68.3  | 62.0  | 61.2  | 67.6  | 71.6  | 74.8  | 75.4  | 79.0  |
|                                                                        | 10%         | WB     | 69.9  | 69.8  | 67.9  | 63.9  | 58.2  | 52.5  | 52.1  | 56.0  | 60.1  | 63.2  | 65.9  | 68.7  |
|                                                                        |             | MCDB   | 77.4  | 76.3  | 75.1  | 70.5  | 66.2  | 61.0  | 59.8  | 64.7  | 68.7  | 71.7  | 73.4  | 75.9  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 15.9  | 15.9  | 17.0  | 18.7  | 22.9  | 22.9  | 23.3  | 22.8  | 21.2  | 18.0  | 15.6  | 15.6  |
|                                                                        |             | MCDBR  | 24.4  | 22.5  | 24.2  | 25.7  | 29.2  | 27.7  | 29.1  | 31.7  | 32.8  | 29.9  | 27.5  | 24.6  |
|                                                                        |             | MCWBR  | 10.1  | 9.5   | 9.9   | 9.8   | 12.4  | 12.2  | 12.5  | 12.9  | 12.5  | 12.0  | 11.9  | 12.0  |
|                                                                        | 5% WB       | MCDBR  | 21.4  | 20.3  | 20.5  | 21.1  | 24.1  | 24.5  | 25.1  | 25.5  | 25.1  | 23.9  | 21.3  | 22.7  |
|                                                                        |             | MCWBR  | 10.5  | 9.7   | 10.3  | 9.7   | 11.7  | 12.9  | 12.8  | 11.7  | 11.6  | 11.5  | 11.1  | 12.1  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.399 | 0.396 | 0.382 | 0.348 | 0.322 | 0.305 | 0.314 | 0.371 | 0.416 | 0.425 | 0.405 | 0.400 |
|                                                                        | taud        |        | 2.356 | 2.363 | 2.383 | 2.440 | 2.473 | 2.496 | 2.461 | 2.308 | 2.198 | 2.216 | 2.303 | 2.343 |
|                                                                        | Ebn at Noon |        | 299   | 294   | 287   | 281   | 274   | 272   | 273   | 269   | 272   | 282   | 295   | 300   |
|                                                                        | Edh at Noon |        | 42    | 41    | 37    | 32    | 28    | 25    | 28    | 36    | 44    | 46    | 44    | 43    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1870  | 1809  | 1597  | 1332  | 1138  | 1020  | 1096  | 1277  | 1487  | 1611  | 1767  | 1842  |
|                                                                        | RadStd      |        | 147   | 143   | 85    | 84    | 61    | 53    | 65    | 75    | 91    | 144   | 134   | 121   |

## Historical Trends

|                             | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only           | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (9 neighbors) | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page